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North-west plastics Manufacturing (Klerksdorp)

FINAL REPORT

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Introduction

This technical report presents the design, implementation and testing of a secure network for North-West Plastics Manufacturing using Cisco Packet Tracer. The purpose of the project was to develop a network infrastructure that supports wired computers, wireless laptops, a server, legacy networking equipment and Internet of Things (IoT) devices. The network was designed using VLANs to separate devices according to their functions.

Four VLANs were implemented which are :

- Main LAN (VLAN 10)
- Legacy Devices (VLAN 20)
- IoT (VLAN 30)
- Network Management (VLAN 40)

The implementation includes:

- router-on-a-stick inter-VLAN routing
- static IP addressing
- WPA2-PSK wireless authentication

Network connectivity was evaluated using ping commands, switch and router verification commands, and Address Resolution Protocol (ARP) table inspection. This report describes the network architecture, configuration procedures, testing results, troubleshooting activities and the extent to which the project requirements were achieved.

Client Requirements

Functional Requirements

The client expects a functional network where :

- All required devices are connected.
- Routers, switches and end devices are configured.
- The network passes connectivity tests.

Cisco Packet Tracer must be used for implementation.

Security Requirements and Constraints

- The wireless network must use WPA2-PSK
- A secure pre-shared password or key must be configured.
- The IoT sensors must be placed on an isolated network segment.
- Legacy devices without modern security features must still be accommodated safely.
- The WPA2-PSK configuration must be tested.
- Any security related troubleshooting must be documented.

Change request Requirements

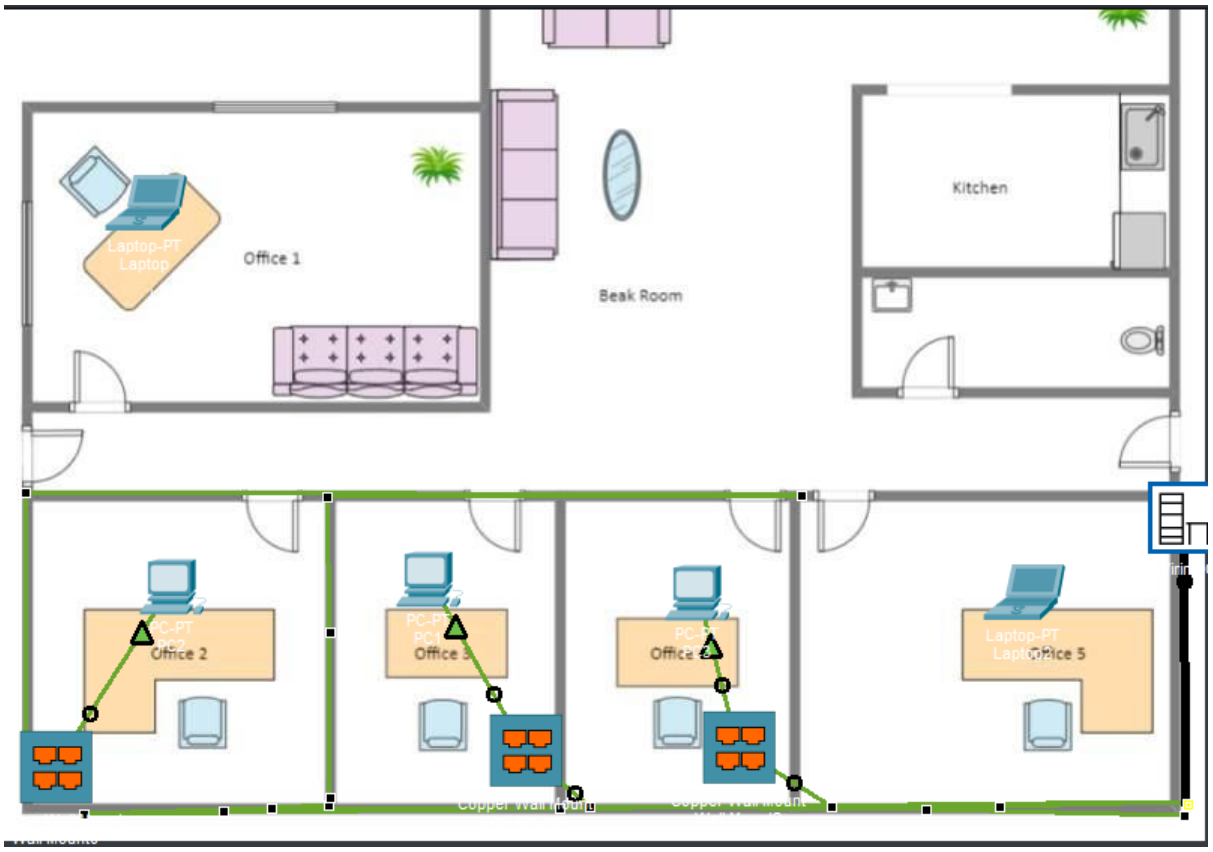
- The network must include IoT sensors.
- A separate network segment must be created.
- Network configuration must be included so that IoT devices can communicate as required.

Performance and reliability requirements

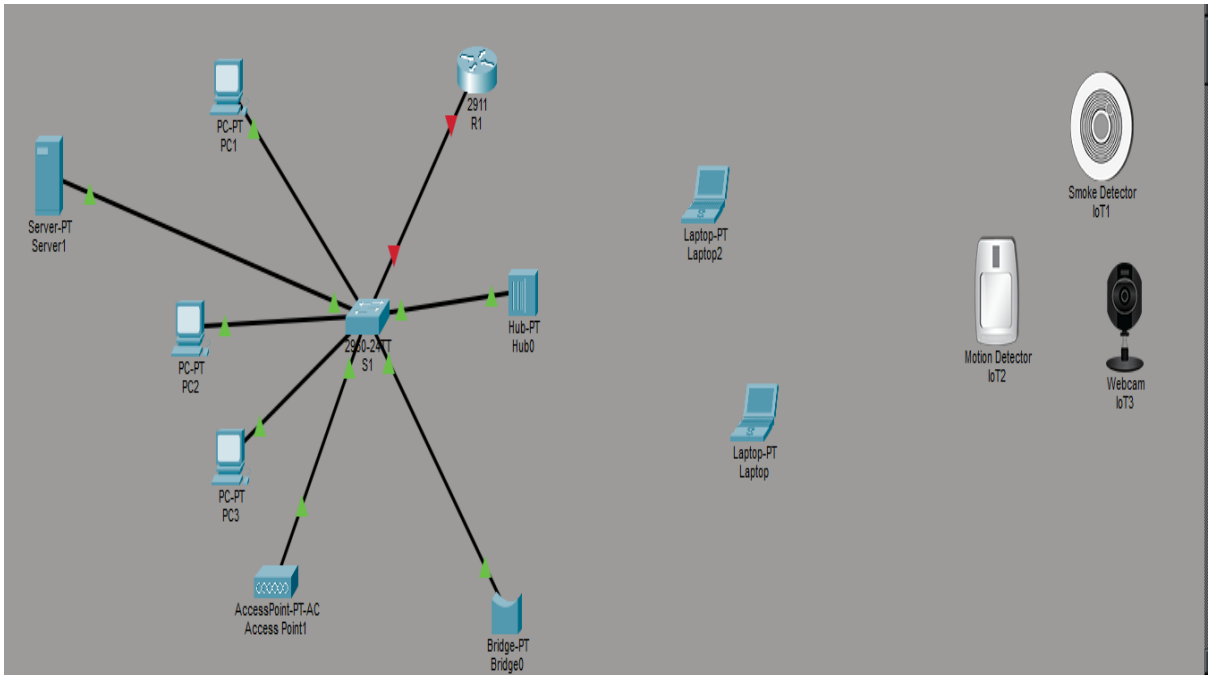
- Relevant devices must be tested for connectivity
- Networking configuration must be verified if it works
- Important troubleshooting recorded must be documented
- The assigned networking challenge must be verified
- The Packet Tracer file must open and work correctly

Topologies

Physical Topology:



Logical Topology



IP Addressing Plan

IP Addressing Block: 192.168.23.0/24

Subnet Mask :255.255.255.0

The /24 addressing block is divided into smaller sub nets used to separate the different network segments since the client change request required that the IoT sensors must be added to the operations area and need an isolated segment.

VLAN	Network Purpose	Network Address	Subnet Mask	Default Gateway	Usable Host Range
VLAN 10	Main LAN	192.168.23.0/26	255.255.255.192	192.168.23.1	192.168.23.2-192.168.23.62
VLAN 20	Legacy Devices	192.168.23.64/27	255.255.255.224	192.168.23.65	192.168.23.66-192.168.23.94
VLAN 30	IoT sensors(isolated)	192.168.23.96/27	255.255.255.224	192.168.23.97	192.168.23.98-192.168.23.126
VLAN 40	Network Management	192.168.23.128/28	255.255.255.240	192.168.23.129	192.168.23.130-192.168.23.142

Device	VLAN	IP Adress	Subnet Mask	Gateway
RouterG0/0.10	10	192.168.23.1	255.255.255.192	
RouterG0/0.20	20	192.168.23.65	255.255.255.224	
RouterG0/0.30	30	192.168.23.97	255.255.255.224	
RouterG0/0.40	40	192.168.23.129	255.255.255.240	
PC1	10	192.168.23.10	255.255.255.192	192.168.23.1
PC2	10	192.168.23.11	255.255.255.192	192.168.23.1
Laptop1	10	192.168.23.12	255.255.255.192	192.168.23.1
Laptop2	10	192.168.23.13	255.255.255.192	192.168.23.1
API	10	192.168.23.20	255.255.255.192	192.168.23.1
Legacy Device1	20	192.168.23.70	255.255.255.224	192.168.23.65
Legacy Device2	20	192.168.23.71	255.255.255.224	192.168.23.65
IoT sensor1	30	192.168.23.100	255.255.255.224	192.168.23.97
IoT sensor2	30	192.168.23.101	255.255.255.224	192.168.23.97
IoT sensor 3	39	192.168.23.102	255.255.255.224	192.168.23.97
Switch Management	40	192.168.23.130	255.255.255.240	192.168.23.129
Server	10	192.168.23.30	255.255.255.192	192.168.23.1

Configuring And Testing Evidence

VLAN configuration on Switch S1

Switch S1 was configured with four VLANs to separate the different sections of the North-West Plastics Manufacturing network. Each VLAN was assigned a specific purpose and IP subnet to organise network traffic and separate devices according to their functions.

```
Switch>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   MAIN_LAN                         active    Fa0/2, Fa0/3, Fa0/4
20   LEGACY_DEVICES                   active    Fa0/6, Fa0/7, Fa0/8
30   IOT                              active    Fa0/5, Fa0/9
40   MANAGEMENT                       active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch>
```

The switch ports were assigned to their respective VLANs. The configuration was verified using the **“show vlan brief “** command which displays the VLANs and their assigned switch ports.

Trunk Port configuration

The **“show interface trunk”** command was used to verify the trunk connection:

```
Switch>show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       10,20,30,40

Port        Vlans allowed and active in management domain
Fa0/1       10,20,30,40

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20,30,40

Switch>
```

Router Sub interface Configuration

Router R1 was configured using the router-on-a-stick method to enable communication between the four VLANs

The sub interface configuration showing the IP addresses and operational status of all four VLAN gateways:

```
Switch>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual up                    up
FastEthernet0/3        unassigned      YES manual up                    up
FastEthernet0/4        unassigned      YES manual up                    up
FastEthernet0/5        unassigned      YES manual up                    up
FastEthernet0/6        unassigned      YES manual up                    up
FastEthernet0/7        unassigned      YES manual up                    up
FastEthernet0/8        unassigned      YES manual up                    up
FastEthernet0/9        unassigned      YES manual down                  down
FastEthernet0/10       unassigned      YES manual down                  down
FastEthernet0/11       unassigned      YES manual down                  down
FastEthernet0/12       unassigned      YES manual down                  down
FastEthernet0/13       unassigned      YES manual down                  down
FastEthernet0/14       unassigned      YES manual down                  down
FastEthernet0/15       unassigned      YES manual down                  down
FastEthernet0/16       unassigned      YES manual down                  down
FastEthernet0/17       unassigned      YES manual down                  down
FastEthernet0/18       unassigned      YES manual down                  down
FastEthernet0/19       unassigned      YES manual down                  down
FastEthernet0/20       unassigned      YES manual down                  down
FastEthernet0/21       unassigned      YES manual down                  down
FastEthernet0/22       unassigned      YES manual down                  down
FastEthernet0/23       unassigned      YES manual down                  down
FastEthernet0/24       unassigned      YES manual down                  down
GigabitEthernet0/1     unassigned      YES manual down                  down
GigabitEthernet0/2     unassigned      YES manual down                  down
Vlan1                  unassigned      YES manual administratively down down
Vlan40                 192.168.23.130  YES manual up                    up
Switch>
```

IP configuration

IP addresses were assigned to network devices according to their respective VLANs. Each VLAN was allocated a separate subnet with the router's corresponding sub interface serving as its default gateway. This addressing scheme allows devices within the same VLAN to communicate and provides routing between VLANs where permitted:

```
C:\>ipconfig

Wireless0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: FE80::20B:BEFF:FE33:242B
   IPv6 Address.....: ::
   IPv4 Address.....: 192.168.23.13
   Subnet Mask.....: 255.255.255.192
   Default Gateway.....: ::
                                 192.168.23.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: ::
   IPv6 Address.....: ::
   IPv4 Address.....: 0.0.0.0
   Subnet Mask.....: 0.0.0.0
   Default Gateway.....: ::
                                 0.0.0.0

C:\>
```

Figure 1Laptop2

```
C:\>ipconfig

Wireless0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: FE80::2D0:BAFF:FE74:791A
   IPv6 Address.....: ::
   IPv4 Address.....: 192.168.23.12
   Subnet Mask.....: 255.255.255.192
   Default Gateway.....: ::
                                 192.168.23.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: ::
   IPv6 Address.....: ::
   IPv4 Address.....: 0.0.0.0
   Subnet Mask.....: 0.0.0.0
   Default Gateway.....: ::
                                 0.0.0.0
```

Figure 2Laptop1

Main LAN Connectivity Test

Connectivity tests were conducted to verify communication between devices on VLAN10 (MAIN_LAN), using the 192.168.23.0/25 subnet. Ping Commands were used to test communication between the laptops, the router's default gateway (192.168.23.1) and Server 1(192.168.23.30).

```
C:\>ping 192.168.23.30

Pinging 192.168.23.30 with 32 bytes of data:

Reply from 192.168.23.30: bytes=32 time=192ms TTL=128
Reply from 192.168.23.30: bytes=32 time=40ms TTL=128
Reply from 192.168.23.30: bytes=32 time=6ms TTL=128
Reply from 192.168.23.30: bytes=32 time=30ms TTL=128

Ping statistics for 192.168.23.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 6ms, Maximum = 192ms, Average = 67ms

C:\>
```

Figure 3Laptop2

```
C:\>ping 192.168.23.1

Pinging 192.168.23.1 with 32 bytes of data:

Reply from 192.168.23.1: bytes=32 time=161ms TTL=255
Reply from 192.168.23.1: bytes=32 time=10ms TTL=255
Reply from 192.168.23.1: bytes=32 time=40ms TTL=255
Reply from 192.168.23.1: bytes=32 time=72ms TTL=255

Ping statistics for 192.168.23.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 10ms, Maximum = 161ms, Average = 70ms

C:\>ping 192.168.23.30

Pinging 192.168.23.30 with 32 bytes of data:

Reply from 192.168.23.30: bytes=32 time=41ms TTL=128
Reply from 192.168.23.30: bytes=32 time=33ms TTL=128
Reply from 192.168.23.30: bytes=32 time=42ms TTL=128
Reply from 192.168.23.30: bytes=32 time=66ms TTL=128

Ping statistics for 192.168.23.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 33ms, Maximum = 66ms, Average = 45ms

C:\>
```

Figure 4Laptop1

IoT VLAN Gateway Connectivity Test

Connectivity tests were conducted to verify communication between router R1 and the three IoT devices In VLAN 30. The IoT network was configured using the 192.168.23.96/27 subnet, with the router sub interface GigabitEthernet0/0.30 assigned the default gateway address 192.168.23.97

Ping commands were executed from the router to each IoT device to verify network connectivity. All three devices responded successfully confirming that they are reachable through AccessPoint1 and the IoT VLAN.

```
Router#ping 192.168.23.102

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.23.102, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 21/39/49 ms

Router#ping 192.168.23.100

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.23.100, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 14/35/74 ms

Router#ping 192.168.23.101

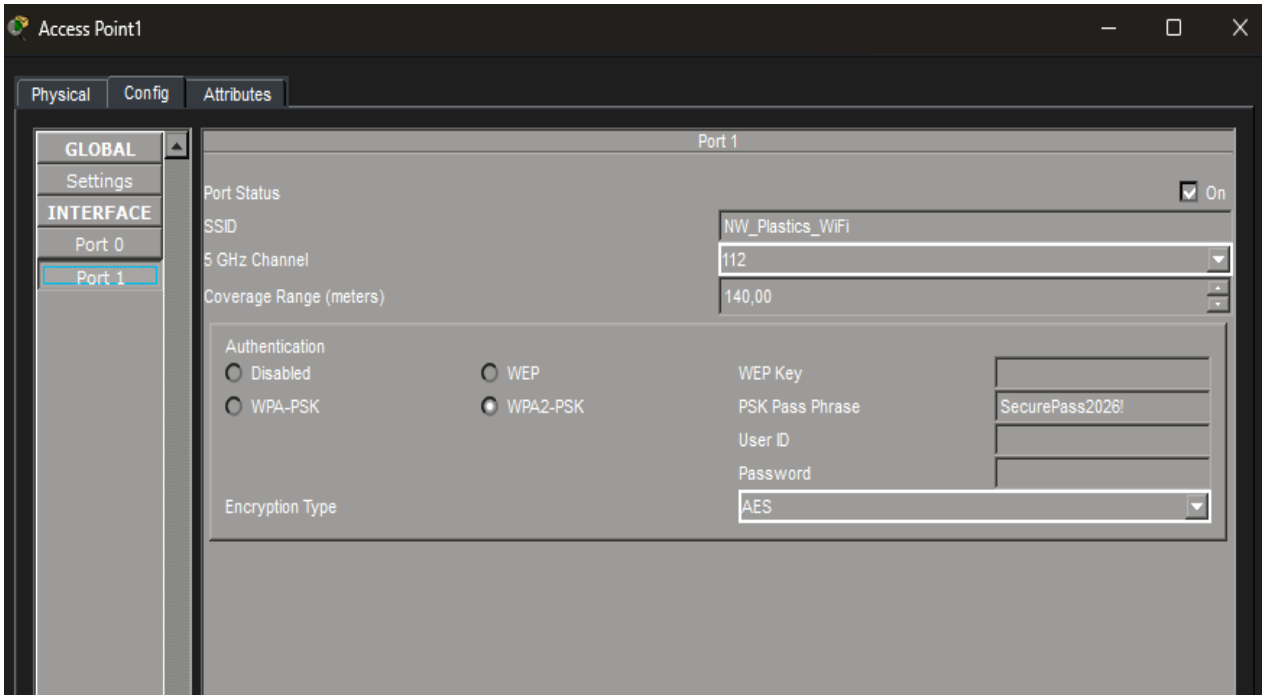
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.23.101, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 14/19/34 ms

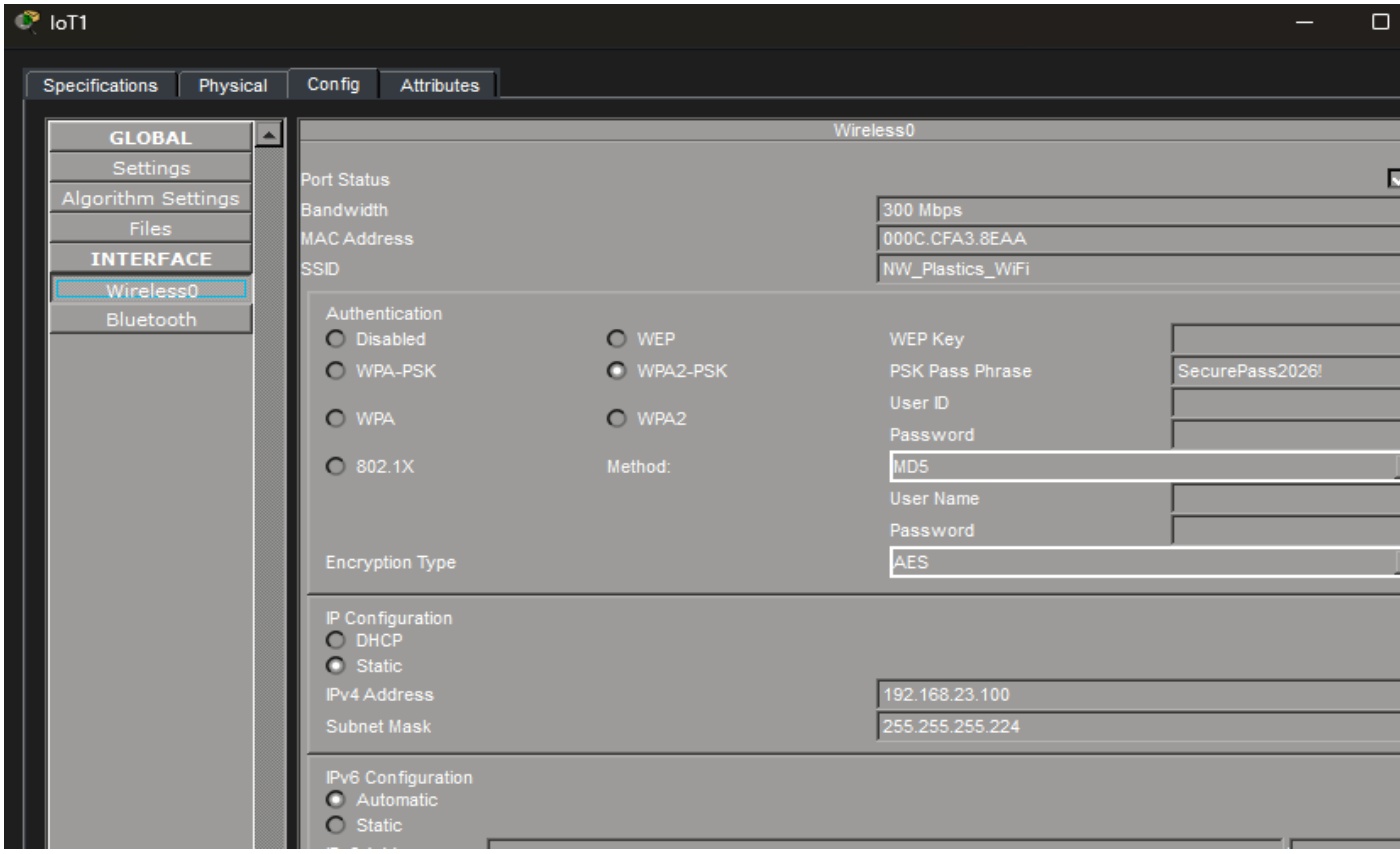
Router#
```

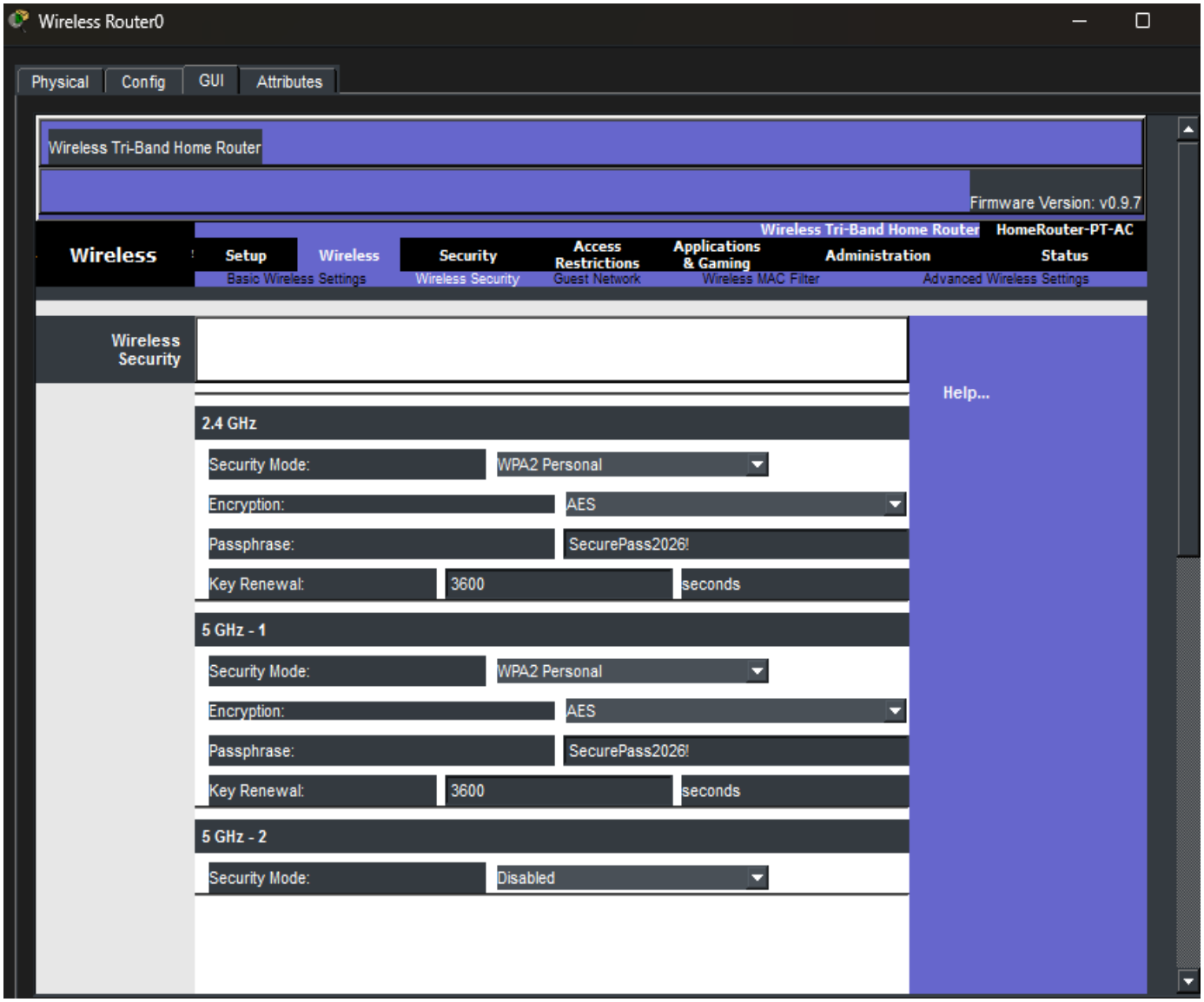
Figure 5 successful connectivity tests between the VLAN 30 gateway and the three IoT sensors

WPA2-PSK Access Point Configuration

WPA2-PSK with AES Encryption was configured on the wireless access points to protect wireless communication. The IoT sensors connect to Access Point 1 in VLAN 30, while the laptops connect to the MAIN_LAN wireless Network in VLAN 10.

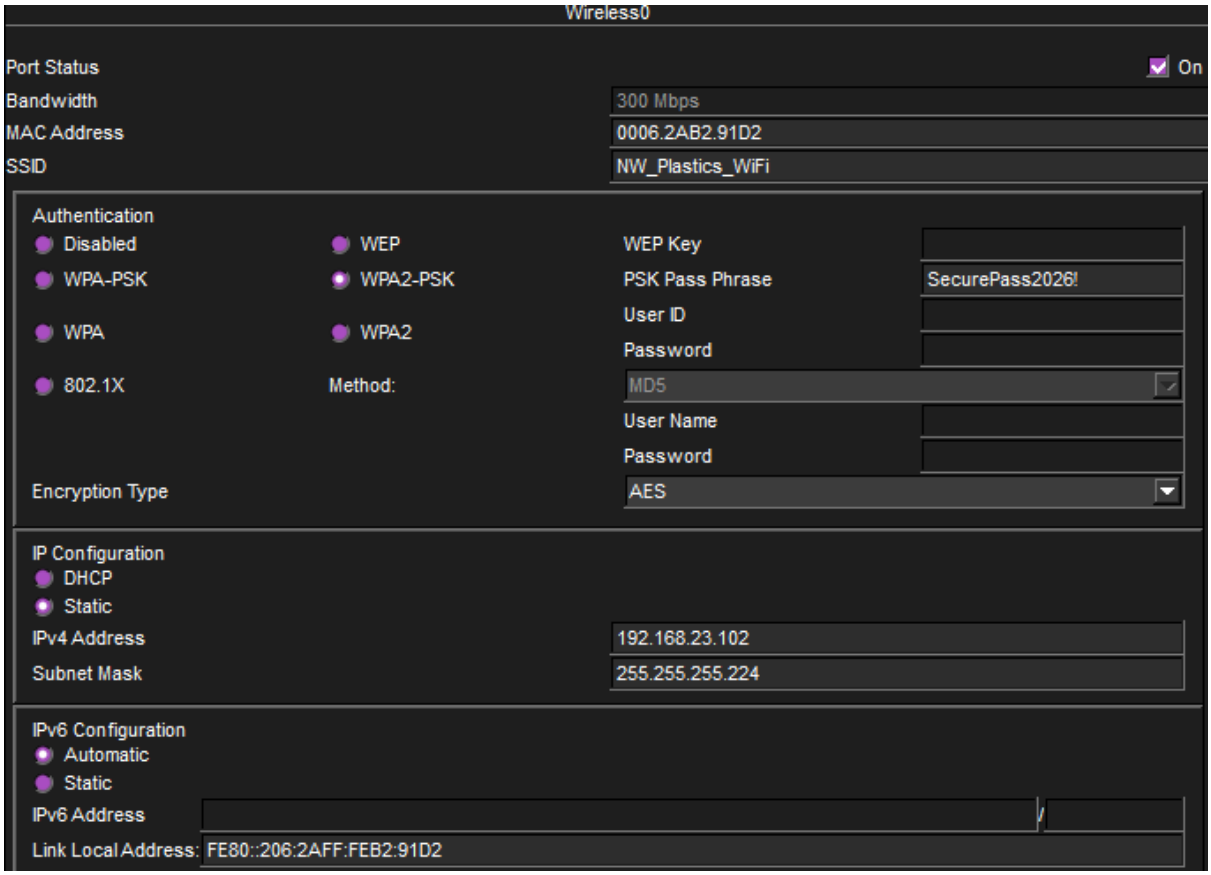


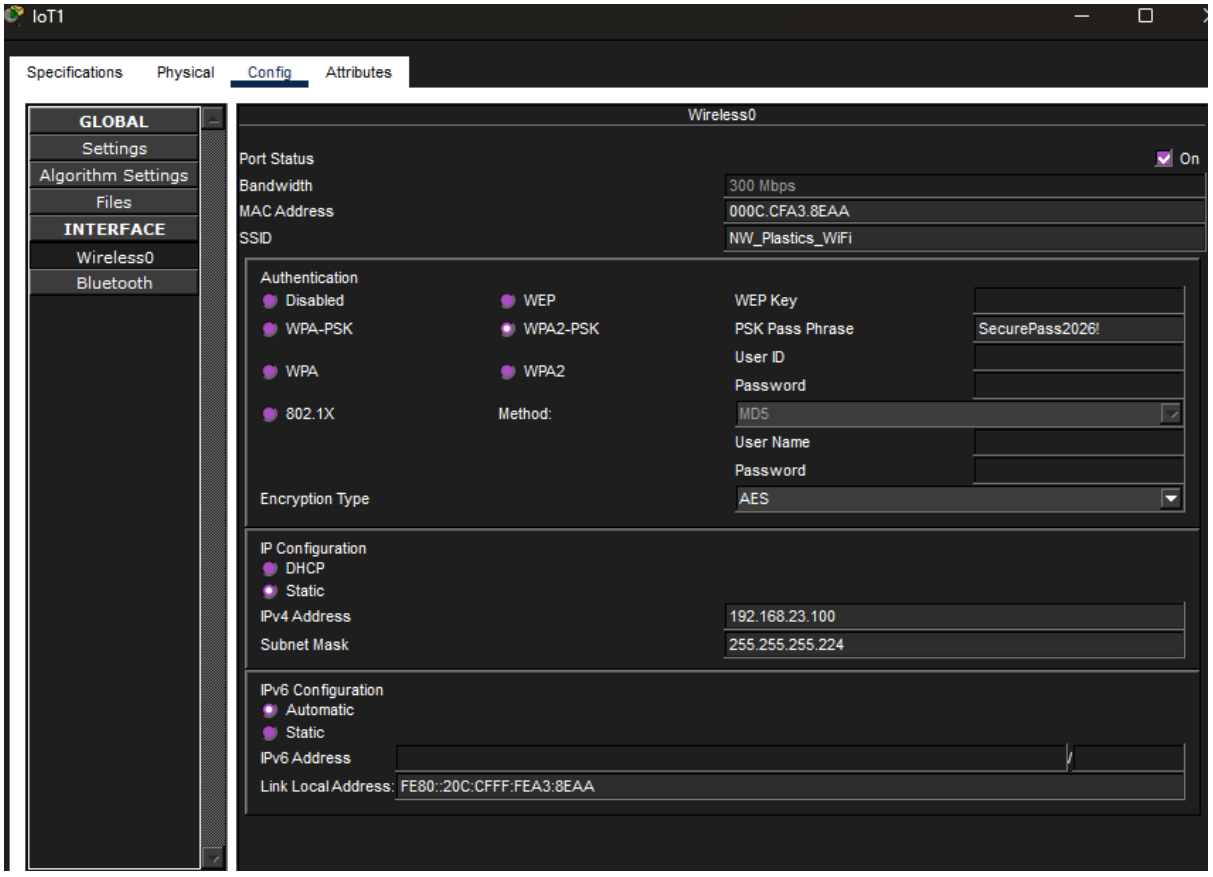

IoT1
Specifications
Physical
Config
Attributes
GLOBAL
Settings
Algorithm Settings
Files
INTERFACE
Wireless0
Bluetooth
Wireless0
Port Status
Bandwidth
300 Mbps
MAC Address
000C.CFA3.8EAA
SSID
NW_Plastics_WiFi
Authentication
Disabled
WEP
WEP Key
WPA-PSK
WPA2-PSK
PSK Pass Phrase
SecurePass2026!
User ID
WPA
WPA2
Password
802.1X
Method:
MD5
User Name
Password
Encryption Type
AES
IP Configuration
DHCP
Static
IPv4 Address
192.168.23.100
Subnet Mask
255.255.255.224
IPv6 Configuration
Automatic
Static

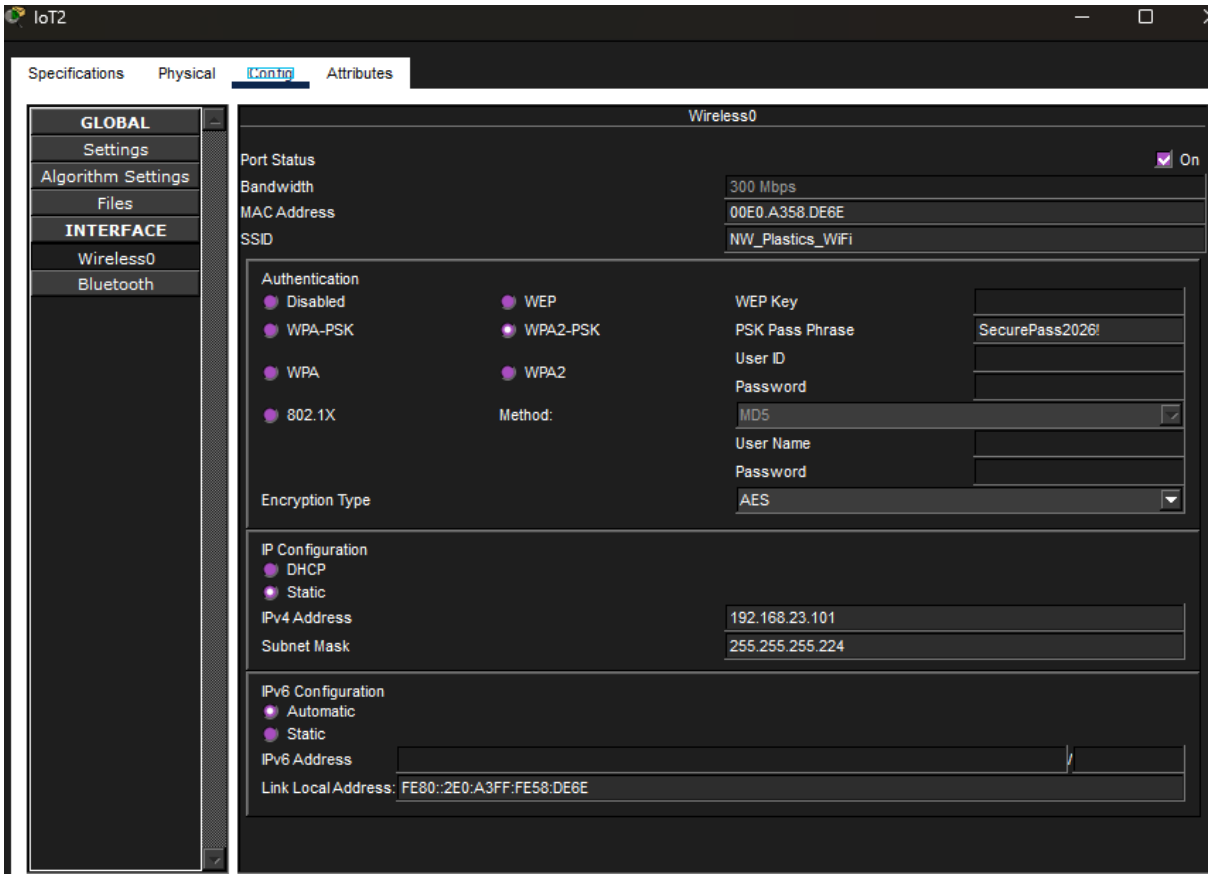

Wireless Router0
Physical
Config
GUI
Attributes
Wireless Tri-Band Home Router
Firmware Version: v0.9.7
Wireless Tri-Band Home Router
HomeRouter-PT-AC
Wireless
Setup
Wireless
Security
Access Restrictions
Applications & Gaming
Administration
Status
Basic Wireless Settings
Wireless Security
Guest Network
Wireless MAC Filter
Advanced Wireless Settings
Wireless Security
Help...
2.4 GHz
Security Mode:
WPA2 Personal
Encryption:
AES
Passphrase:
SecurePass2026!
Key Renewal:
3600
seconds
5 GHz - 1
Security Mode:
WPA2 Personal
Encryption:
AES
Passphrase:
SecurePass2026!
Key Renewal:
3600
seconds
5 GHz - 2
Security Mode:
Disabled

IoT Device Wireless and IP configuration

The three IoT devices were configured to connect wirelessly to AccessPoint1 in VLAN 30. WPA2-PSK authentication was used to secure their wireless connections, with each device configures to use the same SSID and wireless password as the access point.




IoT1
Specifications
Physical
Config
Attributes
GLOBAL
Settings
Algorithm Settings
Files
INTERFACE
Wireless0
Bluetooth
Wireless0
Port Status
On
Bandwidth
300 Mbps
MAC Address
000C.CFA3.8EAA
SSID
NW_Plastics_WiFi
Authentication
Disabled
WEP
WPA-PSK
WPA2-PSK
WPA
WPA2
802.1X
Method:
MD5
WEP Key
PSK Pass Phrase
SecurePass2026!
User ID
Password
User Name
Password
Encryption Type
AES
IP Configuration
DHCP
Static
IPv4 Address
192.168.23.100
Subnet Mask
255.255.255.224
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address:
FE80::20C:CFFF:FEA3:8EAA


IoT2
Specifications
Physical
Config
Attributes
GLOBAL
Settings
Algorithm Settings
Files
INTERFACE
Wireless0
Bluetooth
Wireless0
Port Status
On
Bandwidth
300 Mbps
MAC Address
00E0.A358.DE6E
SSID
NW_Plastics_WiFi
Authentication
Disabled
WEP
WPA-PSK
WPA2-PSK
WPA
WPA2
802.1X
Method:
MD5
WEP Key
PSK Pass Phrase
SecurePass2026!
User ID
Password
User Name
Password
Encryption Type
AES
IP Configuration
DHCP
Static
IPv4 Address
192.168.23.101
Subnet Mask
255.255.255.224
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address:
FE80::2E0:A3FF:FE58:DE6E

IoT ARP Table Verification

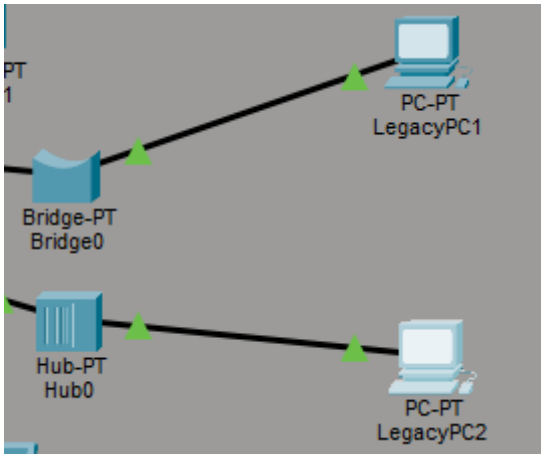
The router's ARP table was inspected to verify that it had learned the MAC addresses associated with the three IoT devices. The entries for 192.168.23.100, 192.168.23.101, 192.168.23.102 were checked on interface GigabitEthernet0/0.30

```
Router>enable
Router#show ip arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.23.13           40  00D0.BA74.791A  ARPA   GigabitEthernet0/0.10
Internet  192.168.23.14           38  00D0.BA74.791A  ARPA   GigabitEthernet0/0.10
Internet  192.168.23.100          22  000C.CFA3.8EAA  ARPA   GigabitEthernet0/0.30
Internet  192.168.23.101          22  00E0.A358.DE6E  ARPA   GigabitEthernet0/0.30
Internet  192.168.23.102          31  000C.CFA3.8EAA  ARPA   GigabitEthernet0/0.30
Router#
```

Figure 6 Router ARP table

Legacy VLAN Configuration and Testing

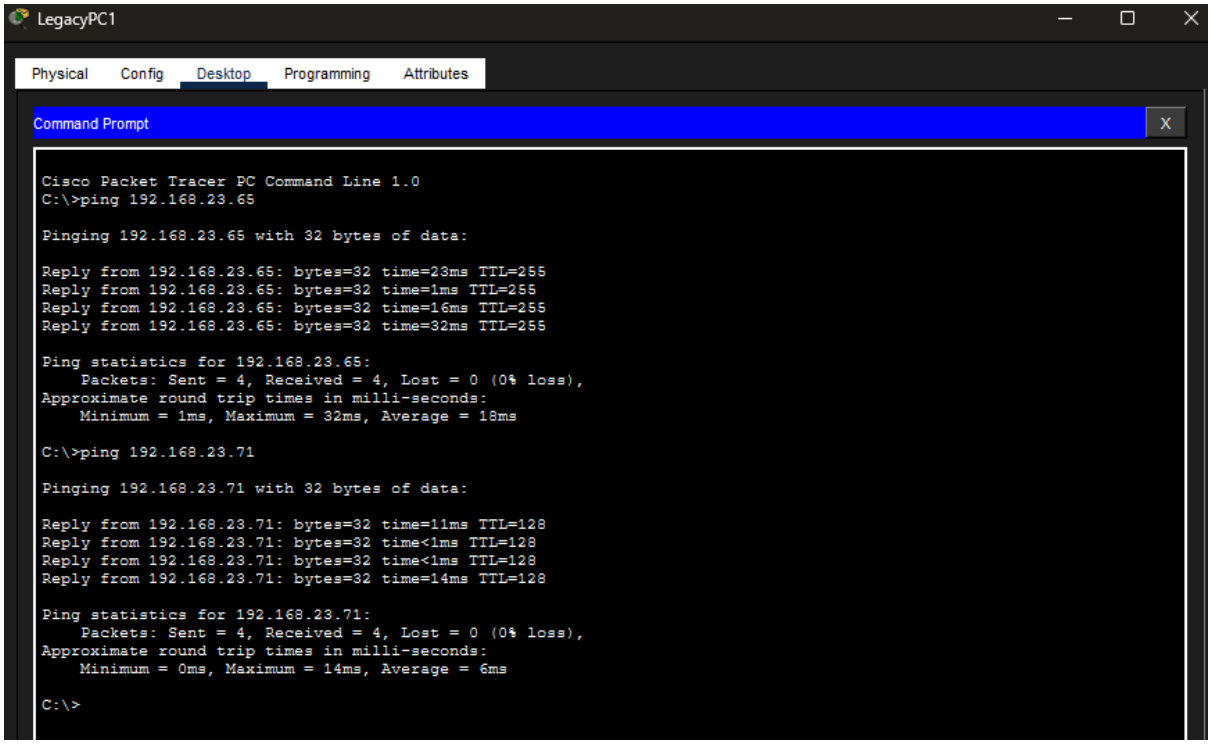
To test that the legacy devices can communicate through VLAN 20, Legacy PC1 and PC2 were connected through a Hub and a Bridge as shown in the picture :



Both PCs have a subnet mask of 255.255.255.224 and gateway 192.168.23.65

PC1 has IP address 192.168.23.70 then PC2 has IP address 192.168.23.71.

The connectivity Testing is in the image below:



This is evident that the Legacy devices communicate though VLAN20.

Network Management VLAN Configuration

VLAN40 was configured for network management using subnet 192.168.23.128/28.The router's management gateway was assigned 192.168.23.130.

```
Switch>enable
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/1, Gig0/2
10   MAIN_LAN                         active    Fa0/2, Fa0/3, Fa0/4, Fa0/8
                                                Fa0/10
20   LEGACY_DEVICES                   active    Fa0/6, Fa0/7
30   IOT                              active    Fa0/5, Fa0/9
40   MANAGEMENT                       active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

```
Switch#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual up                    up
FastEthernet0/3        unassigned      YES manual up                    up
FastEthernet0/4        unassigned      YES manual up                    up
FastEthernet0/5        unassigned      YES manual up                    up
FastEthernet0/6        unassigned      YES manual up                    up
FastEthernet0/7        unassigned      YES manual up                    up
FastEthernet0/8        unassigned      YES manual up                    up
FastEthernet0/9        unassigned      YES manual down                  down
FastEthernet0/10       unassigned      YES manual up                    up
FastEthernet0/11       unassigned      YES manual down                  down
FastEthernet0/12       unassigned      YES manual down                  down
FastEthernet0/13       unassigned      YES manual down                  down
FastEthernet0/14       unassigned      YES manual down                  down
FastEthernet0/15       unassigned      YES manual down                  down
FastEthernet0/16       unassigned      YES manual down                  down
FastEthernet0/17       unassigned      YES manual down                  down
FastEthernet0/18       unassigned      YES manual down                  down
FastEthernet0/19       unassigned      YES manual down                  down
FastEthernet0/20       unassigned      YES manual down                  down
FastEthernet0/21       unassigned      YES manual down                  down
FastEthernet0/22       unassigned      YES manual down                  down
FastEthernet0/23       unassigned      YES manual down                  down
FastEthernet0/24       unassigned      YES manual down                  down
GigabitEthernet0/1     unassigned      YES manual down                  down
GigabitEthernet0/2     unassigned      YES manual down                  down
Vlan1                  unassigned      YES manual administratively down down
Vlan40                 192.168.23.130  YES manual up                    up
Switch#
```

Connectivity was verified using ping tests between the router and switch:

```
Router>enable
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.23.1    YES manual up                    up
GigabitEthernet0/0.20  192.168.23.65   YES manual up                    up
GigabitEthernet0/0.30  192.168.23.97   YES manual up                    up
GigabitEthernet0/0.40  192.168.23.129  YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#ping 192.168.23.130

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.23.130, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router#
```

```
Switch#show interface trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       10,20,30,40

Port        Vlans allowed and active in management domain
Fa0/1       10,20,30,40

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20,30,40

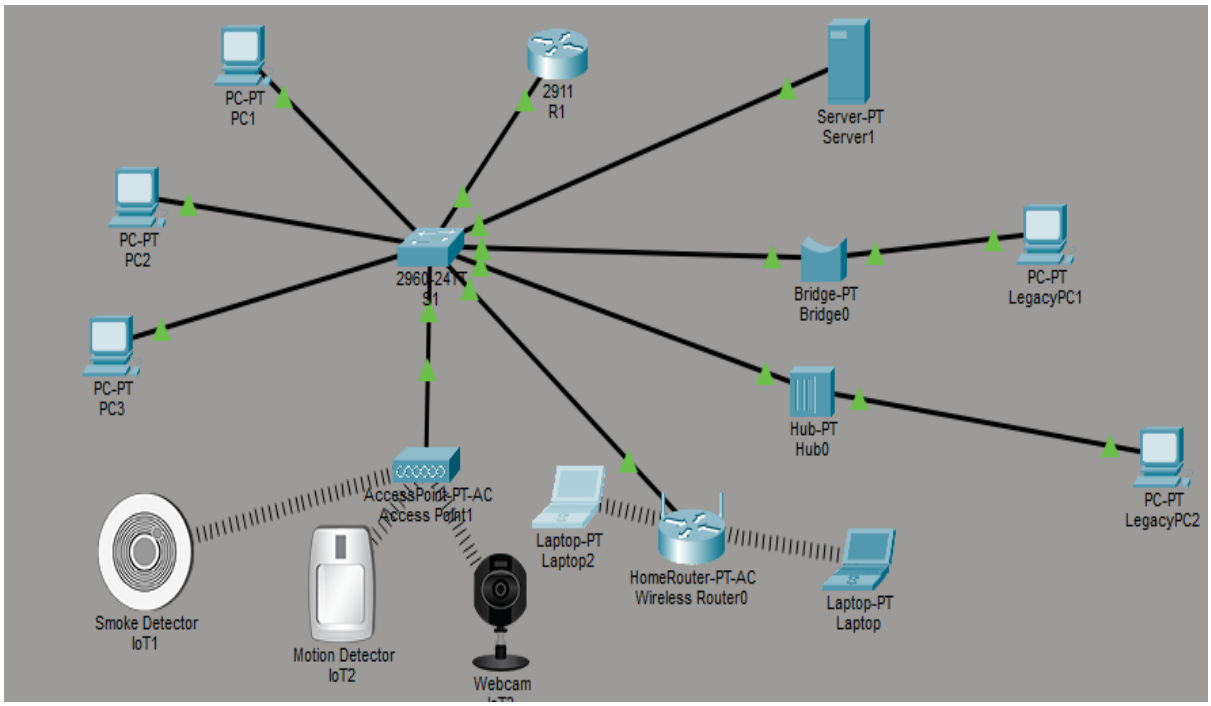
Switch#ping 192.168.23.129

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.23.129, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/14/41 ms

Switch#
```

Final Topology

After configuration, all devices as well as switches and routers were assigned to their correct connections, and the network was working properly :



Troubleshooting

During the network implementation a connectivity problem occurred between Laptop2 and Server1 in the main LAN. Initial ping tests showed that laptop2 could communicate with the default gateway but attempts to reach Server1 were unsuccessful.

The switch VLAN configuration was inspected to identify the cause of the problem. It was discovered that Serber1 was connected to switchport Fa0/8, which had been assigned to VLAN 20 instead of VLAN 10.

The issue was resolved by reconfiguring Fa0/8 as an access port in VLAN10. After the correction, the connectivity test was repeated, and Laptop2 successfully communicated with Server1, receiving four out of four ping replies.

This troubleshooting process demonstrated the importance of VLAN assignments and systematic connectivity testing.

Conclusion

The Northwest Plastics Manufacturing network project demonstrated the design, configuration and testing of a network using Cisco packet tracer. It focused on providing reliable connectivity also improving network organisation and implementing wireless security while accommodating both modem and legacy networking devices. Overall, the project provided practical experience in network design , VLAN, configuration, wireless security , IP addressing and troubleshooting . It also highlighted the importance of network segmentation and the need for additional access controls to enforce IoT isolation. The knowledge and skills gained from this project can be also applied to the design and maintenance of secure networks in real world environments.

Final GitHub Repository

https://github.com/Almagry/CMPG325-2026-030-Wireless-Security-

Bibliography

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